

ARIA Investment Intelligence Memo

Generated: 18 June 2026 | Persona: Investor | Market: Athens vs Paris

1. Executive Summary

Query: "Compare Paris and Athens for a EUR 500k STR investment. Which market, which neighbourhood, projected annual yield, AMA compliance risk in central Athens, and what host improvements would maximise yield?"

ARIA activated 5 specialist agent(s): pricing, risk, forecast, compliance, coach.

All five prior phase models contributed to this analysis.

Key signal: BUY -- Athens occupancy growing toward 2026-07 peak.

2. Pricing Intelligence (Phase 2 XGBoost -- $R^2=0.676$)

City: Athens

Predicted fair value: EUR 79/night

Current listed price: EUR 78/night

Underpricing gap: EUR 1.1/night (fairly priced)

Top SHAP price drivers: listing_index, neighbourhood_median_price, review_score_composite

Market context: 2,945 Athens listings carry a median EUR 25 gap. Aggregate foregone revenue = EUR 4.8M/yr. The 865 priority listings (underpriced AND Phase 3 high-risk) represent EUR 1.43M concentrated opportunity with two compounding upside sources.

3. Demand Forecast (Phase 4 Prophet -- MAPE=19.3%)

Investor signal: BUY

Mean monthly occupancy: 9.5 days/month

Annual occupied days: 114 days/year

Peak demand month: 2026-07

Revenue scenarios (Athens):

Base scenario: EUR 9,732/yr

Optimistic +10%: EUR 10,705/yr

Downside -15%: EUR 8,272/yr

Paris comparison: BUY signal | EUR 20,041/yr base revenue

4. Host Risk and Supply Dynamics (Phase 3 LightGBM -- AUC=0.8288)

Risk band: MEDIUM

Risk probability: 0.531

Action: MONITOR -- moderate risk. Review review_velocity_130d monthly.

Priority listings: 865 (underpriced AND high-risk)

Supply-shock thesis: High-risk hosts who exit release their demand to surviving compliant operators in the same neighbourhood. Combined with Phase 5 enforcement removal of 137 unlicensed listings (EUR 991k/yr at market rates), two independent supply-reduction mechanisms compound the opportunity for compliant, correctly-priced operators.

5. Regulatory Compliance (Phase 5 RAG -- Phase 5 RAG (live))

Framework: AMA Law 4276/2014 + Dec 2024 freeze

AMA freeze applies: False

Market compliance rate: 99.0%

Unlicensed (market): 137 listings

HIGH risk: 27 (cannot regularise -- AMA freeze zone)

Revenue at risk: EUR 991,324/yr

Investor note: Central Athens AMA freeze Dec 2024: no new registrations. Licensed properties in freeze zones have protected market position. Market: 99.0% compliant. Enforcement removes 137 unlicensed (EUR 991k/yr) -- demand redistributes.

Citation: AMA Registration Portal - Greek Ministry of Tourism

SHAP-Based Host Recommendations (Phase 2 SHAP Analysis)

* SHAP rank 2 Athens: review_score_composite -- Listings below 9.0 face systematic underpricing. Add wifi, AC, kitchen essentials to reach the 9.0 threshold. Each +0.1 point correlates with ~EUR 3 price uplift.

* SHAP rank 1 both cities: neighbourhood_median_price -- Your neighbourhood sets the price floor. Raise price to neighbourhood median before premium positioning.

ARIA -- Agentic Real-estate Intelligence Advisor

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Human review required before client delivery.